

- 6 Forces 5 N, 12 N and 13 N act at a point which is in equilibrium.

What is the angle between the 5 N and 13 N force?

A 23°

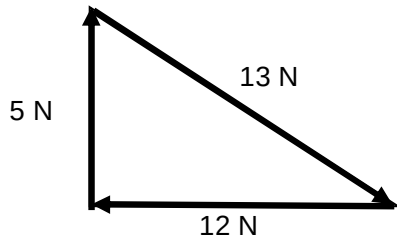
B 67°

C 90°

D 113°

Ans: D

$$\begin{aligned}\text{Angle} &= 180 - \tan^{-1}(12/5) \\ &= 180 - 67 \\ &= 113\end{aligned}$$



7

7 A cube of sides 10 cm has a density of 7.9 g cm^{-3} . It floats vertically with one eighth of its side